MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $u_n = 3n + 2$: u_4 equals
   - A. 14
   - B. 18
   - C. 12
   - D. 6
2. $u_0 = 2$, $u_{n+1} = u_n + 5$: u_3 equals
   - A. 10
   - B. 12
   - C. 17
   - D. 15
3. An increase of 8% gives the ratio
   - A. 0.08
   - B. 1.8
   - C. 0.92
   - D. 1.08
4. From u_3 to u_9 inclusive, there are
   - A. 6
   - B. 7
   - C. 9
   - D. 12
5. $1 + 2 + \cdots + 20$ equals
   - A. 200
   - B. 420
   - C. 210
   - D. 190

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $u_n = -3 \times 2^n$ is
   - A. strictly increasing
   - B. constant
   - C. neither increasing nor decreasing
   - D. strictly decreasing
2. $1 + 2 + 4 + 8$ equals
   - A. 15
   - B. 16

C. 14

D. 8

3. For $u_n = \dfrac{16}{2^n}$, the first index where $u_n < 2$:

   A. 2

   B. 3

   C. 4

   D. 5

4. The sequence $(-1)^n$

   A. tends to 0

   B. has no limit

   C. is increasing

   D. is constant

5. A list contains u_0 to u_{10}; its length is

   A. 10

   B. 9

   C. 12

   D. 11

True or false? Justify.

1. Every geometric sequence with common ratio $q > 1$ is increasing.
2. If two sequences have their first three terms equal, they are equal.
3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

Key point · Self-assessment. I distinguish explicit from recurrence definitions; I recognise a constant increment or rate; I count the terms; I check the previous index of a threshold.

For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.